

VIRGINIA TECH MONTGOMERY, VA, USA

WMO: 724113

Lat: 37.208N

Lon: 80.408W

Elev: 650

StdP: 93.76

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 53881

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -12.0	(c) -9.0	(d) -19.0	(e) 0.8	(f) -8.0	(g) -16.8	(h) 0.9	(i) -7.3	(j) 11.0	(k) -2.3	(l) 9.8	(m) -1.7	(n) 3.5	(o) 300	(p) 0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	31.6	22.4	30.2	21.8	28.9	21.4	24.2	28.6	23.5	27.7	22.8	26.8	2.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.1	25.9	22.4	18.5	25.5	21.7	17.7	24.9	76.9	28.1	73.4	27.7	70.9	26.9	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.8	(b) 7.7	(c) 6.7	(e) -15.1	(f) 33.3	(g) 2.7	(h) 1.5	(i) -17.0	(j) 34.3	(k) -18.6	(l) 35.2	(m) -20.1	(n) 36.0	(o) -22.1	(p) 37.1		
(a) 8.8	(b) 7.7	(c) 6.7	(e) -15.8	(f) 25.0	(g) 2.6	(h) 1.1	(i) -17.7	(j) 25.8	(k) -19.2	(l) 26.4	(m) -20.7	(n) 27.1	(o) -22.6	(p) 27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	1.0	2.7	6.8	12.3	17.0	21.0	22.9	22.3	18.9	13.0	7.0	2.9
	DBStd	8.89	5.84	5.46	5.33	4.68	3.92	2.87	2.36	2.44	3.49	4.41	4.81	5.01
	HDD10.0	1016	286	210	127	29	2	0	0	0	0	22	114	226
	HDD18.3	2628	538	437	357	184	72	10	1	2	35	173	341	477
	CDD10.0	1882	6	6	29	99	220	331	399	383	266	114	24	6
	CDD18.3	454	0	0	0	4	31	91	142	127	51	7	0	0
	CDH23.3	3542	0	1	8	90	279	693	1084	928	393	62	4	0
	CDH26.7	979	0	0	0	13	49	183	344	281	99	9	0	0
Wind	WSAvg	2.2	3.0	3.0	3.0	2.8	2.1	1.7	1.5	1.4	1.6	1.9	2.3	2.5
Precipitation	PrecAvg	1042	76	75	93	90	106	92	106	93	86	79	73	75
	PrecMax	1464	188	160	212	197	248	220	238	190	225	232	188	164
	PrecMin	729	15	9	15	19	23	10	40	22	6	0	10	5
	PrecStd	172	34	39	40	40	50	50	44	36	52	47	40	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.3	20.1	24.1	28.0	29.8	32.4	32.9	32.8	31.3	27.6	22.9	18.9
		MCWB	13.8	14.4	15.4	17.5	19.7	22.8	23.4	22.8	20.1	19.1	15.0	14.8
	2%	DB	15.9	17.2	21.6	26.1	27.8	30.2	31.4	31.3	29.1	25.0	20.0	16.2
		MCWB	12.4	11.3	13.9	16.3	19.4	21.8	22.9	22.4	20.5	17.9	13.5	12.7
	5%	DB	12.8	14.3	18.6	23.7	26.8	28.7	30.0	29.7	27.5	22.7	17.6	13.7
		MCWB	9.6	9.7	12.2	15.2	18.9	21.6	22.3	22.0	20.0	16.6	13.1	11.0
	10%	DB	10.2	12.1	16.2	21.4	24.7	27.4	28.7	27.9	25.9	20.9	15.6	11.3
		MCWB	7.6	8.2	10.9	14.2	18.1	20.8	21.9	21.3	19.5	16.3	11.7	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	15.7	16.8	19.3	22.3	24.8	25.5	25.3	23.6	21.7	17.4	16.1
		MCDB	17.7	18.5	22.1	25.3	26.5	29.5	30.2	28.7	27.2	24.7	20.9	17.7
	2%	WB	12.8	13.0	15.1	17.9	21.1	23.2	24.5	24.0	22.4	19.9	15.5	13.9
		MCDB	15.1	15.9	19.4	23.2	25.7	27.7	29.0	28.2	26.0	22.8	18.3	15.5
	5%	WB	10.4	10.7	13.6	16.6	20.1	22.4	23.6	23.3	21.3	18.2	13.9	11.8
		MCDB	12.5	13.5	17.3	21.6	24.7	26.9	27.9	27.3	24.9	21.2	16.4	13.2
	10%	WB	7.5	8.1	11.9	15.3	19.1	21.6	22.8	22.5	20.4	16.7	12.2	8.8
		MCDB	9.6	11.1	15.0	20.0	23.2	26.0	26.8	26.4	24.1	19.7	14.8	10.6
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.4	11.6	12.7	11.9	11.0	10.7	10.8	11.3	11.9	11.8	9.8
		MCDBR	13.4	15.1	16.1	16.6	14.4	13.1	12.8	13.0	14.0	15.0	15.5	13.7
		MCWBR	9.8	10.5	9.3	8.4	7.0	6.0	5.3	5.5	6.4	8.2	10.0	10.0
	5% WB	MCDBR	11.7	13.0	13.3	13.9	12.1	11.3	10.7	10.8	11.2	11.5	12.1	11.8
		MCWBR	9.7	10.3	8.8	7.8	6.4	5.6	5.1	5.1	5.7	7.1	9.0	9.9
Clear-Sky Solar Irradiance	taub	0.293	0.307	0.332	0.375	0.413	0.445	0.475	0.477	0.378	0.342	0.309	0.293	
	taud	2.509	2.459	2.435	2.335	2.233	2.176	2.122	2.091	2.371	2.464	2.524	2.539	
	Ebn at Noon	904	931	933	905	872	840	811	800	872	876	877	881	
	Edh at Noon	79	94	106	124	140	148	155	157	112	93	77	72	
All-Sky Solar Radiation	RadAvg	2.23	2.94	3.99	5.06	5.58	6.08	5.71	5.22	4.38	3.47	2.59	1.94	
	RadStd	0.19	0.35	0.26	0.41	0.48	0.44	0.36	0.27	0.43	0.42	0.25	0.18	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	+0.48	N/A	N/A	N/A	+0.58	+0.72	N/A	N/A	+147
(47) Regional (40 neighbors)	DBAvg	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+132

Nomenclature: See separate page